

Practical 2 - Logistic regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `lbw.RData` and `insecticide.RData`

Introduction

In the first part of this session we will use a publicly available dataset on birthweights of babies, taken from the book *Applied Logistic Regression* by David W Hosmer and Stanley Lemeshow. Per the book:

“Data were collected as part of a larger study at Baystate Medical Center in Springfield, Massachusetts. This dataset contains information on 189 births to women seen in the obstetrics clinic. Fifty nine of these births were defined as low birth weight [defined as less than 2500 grams].”

In the second part of the session we will use data from a toxicological experiment. Eight groups of insects were exposed for five hours to gaseous carbon disulphide (CS_2) at different concentrations, with the purpose of investigating how the risk of death depends on the dose received.

Aims

The aims of this session are:

- understand how to interpret results from logistic regression models.
- understand how to compare the fits of logistic regression models.

Part A: Investigation into low birthweights in Massachusetts, USA

We will first investigate factors which may be related to the chance of having a low birthweight baby, using the lbw dataset. Begin by loading the data with `load("lbw.RData")` (remember to set your working directory or alternatively add the path/folder to the call to `load()`).

Question 1

Summarise the dataset, checking for missing values. Identify which variable stores our outcome of interest. Also note that the mother's weight is given in lbs, whilst the baby's weight is given in grams.

SOLUTION

```
load("lbw.RData")
summary(lbw)
```

id		low		age		lwt			
Min.	: 4.0	Birthweight >= 2500g:	130	Min.	:14.00	Min.	: 80.0		
1st Qu.:	68.0	Birthweight < 2500g :	59	1st Qu.:	19.00	1st Qu.:	110.0		
Median :	123.0			Median :	23.00	Median :	121.0		
Mean :	121.1			Mean :	23.24	Mean :	129.8		
3rd Qu.:	176.0			3rd Qu.:	26.00	3rd Qu.:	140.0		
Max.	:226.0			Max.	:45.00	Max.	:250.0		
race		smoke		ptl		ht		ui	
White:	96	Nonsmoker:	115	Min.	:0.0000	Min.	:0.00000	Min.	:0.0000
Black:	26	Smoker :	74	1st Qu.:	0.0000	1st Qu.:	0.00000	1st Qu.:	0.0000
Other:	67			Median :	0.0000	Median :	0.00000	Median :	0.0000
				Mean :	0.1958	Mean :	0.06349	Mean :	0.1481
				3rd Qu.:	0.0000	3rd Qu.:	0.00000	3rd Qu.:	0.0000
				Max.	:3.0000	Max.	:1.00000	Max.	:1.0000
ftv		bwt							
Min.	:0.0000	Min.	: 709						
1st Qu.:	0.0000	1st Qu.:	2414						
Median :	0.0000	Median :	2977						
Mean :	0.7937	Mean :	2944						
3rd Qu.:	1.0000	3rd Qu.:	3475						
Max.	:6.0000	Max.	:4990						

There do not appear to be any missing values. The outcome variable of interest is `low`. This is stored as a factor, which we can check using the `str` (structure) function:

```
str(lbw$low)
```

```
Factor w/ 2 levels "Birthweight >= 2500g",...: 1 1 1 1 1 1 1 1 1 1 ...
```

This doesn't show the two levels well, because of the length of the string for the first level. To see all the levels, we can use `levels`:

```
levels(lbw$low)
```

```
[1] "Birthweight >= 2500g" "Birthweight < 2500g"
```

Question 2

Use the `glm` function to fit a logistic model for the low birthweight variable with the mother's weight at last menstrual period (in lbs) (`lwt`) as the single covariate.

Convert the estimates and their confidence intervals to odds and odds ratios, and discuss their interpretation with your colleagues.

SOLUTION

```
lwtMod <- glm(low ~ lwt, data=lbw,  
              family="binomial")  
summary(lwtMod)
```

Call:

```
glm(formula = low ~ lwt, family = "binomial", data = lbw)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.995763	0.785243	1.268	0.2048
lwt	-0.014037	0.006169	-2.276	0.0229 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	234.67	on 188	degrees of freedom
Residual deviance:	228.71	on 187	degrees of freedom

AIC: 232.71

Number of Fisher Scoring iterations: 4

```
exp(coef(lwtMod))
```

	(Intercept)	lwt
	2.7067899	0.9860609

```
confint.default(lwtMod)
```

		2.5 %	97.5 %
(Intercept)	-0.54328547	2.534812248	
lwt	-0.02612721	-0.001947035	

```
exp(confint.default(lwtMod))
```

		2.5 %	97.5 %
(Intercept)	0.5808368	12.6140623	
lwt	0.9742112	0.9980549	

0.9957634: the estimated log odds of a low birthweight baby when mother's weight is zero (an extrapolation).

It can be converted to the estimated odds of a low birthweight baby when mother's weight is zero by taking its exponent ($e^{0.9957634} = 2.706788$).

This can be converted to the estimated probability of a low birthweight baby when mother's weight is zero using the standard conversion formula. Recall that $odds = p/(1 - p)$ where p is the probability of the event. Re-arranging this, we have

$$p = \frac{odds}{odds + 1}$$

Here this estimated probability is 0.730 ($2.706788/(2.706788 + 1)$). Since it is an extrapolation to an impossible mother's weight this (and its 95% CI) is meaningless from a practical perspective. It is reported in the output because in order to define any straight line we require the slope of the line and one point on the line.

The reported p-value for the constant term relates to a test of the null hypothesis that log odds of a low birthweight baby equals zero, i.e., that the probability equals 0.5. There is no practical interest in this null hypothesis or the p-value.

-0.0140371: the estimated increase in the log odds of a low birthweight baby per 1lb increase in mother's weight at last menstrual period.

It can be converted to the estimated odds ratio (of having a low birthweight baby) per 1 lb increase in mother's weight by taking its exponent ($e^{-0.0140371} = 0.9860609$). Subtracting 1 from this estimate gives -0.0139391, so we might report that the odds of a low birthweight baby is reduced by an estimated 1.4% per 1lb increase in mother's weight at last menstrual period.

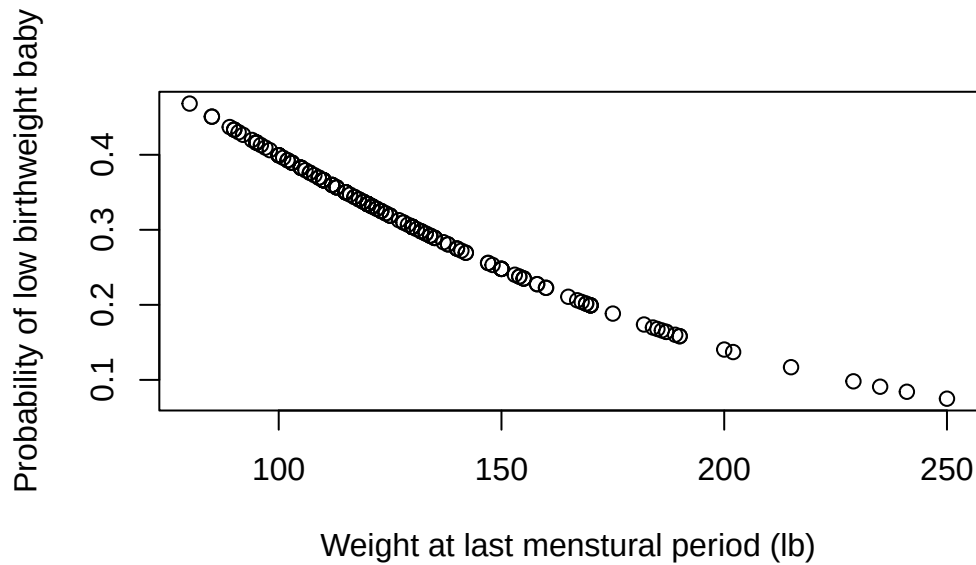
The reported p-value for the slope relates to a test of the null hypothesis that the log odds ratio per 1lb increase in mother's weight is zero, i.e., that the odds ratio equals 1. Here we have evidence ($p=0.023$) that the odds (and therefore the probability) of a low birthweight baby is associated with mother's weight.

The reported 95% CI for the slope relates to the log odds ratio per 1lb increase in mother's weight. We would usually convert the confidence limits to confidence limits for the odds ratio by taking exponents ($e^{-0.0261272} = 0.9742112$, $e^{-0.001947} = 0.9980549$). The interpretation of the 95% CI is the standard one. We can either say that there is a 95% probability that a 95% CI constructed in this way has a 95% probability of containing the true (population) odds ratio, or that the results are consistent ($p > 0.05$) with true odds ratios between 0.974 and 0.998 and inconsistent ($p < 0.05$) with true odds ratios outside this range.

We could summarise the important results here as "There is evidence ($p = 0.023$) of an association between mother's weight and the probability of a low birthweight baby. Odds of a low birthweight baby were estimated to be reduced by 1.4% (95% CI 0.2% to 2.6%) per 1lb increase in mother's weight at last menstrual period".

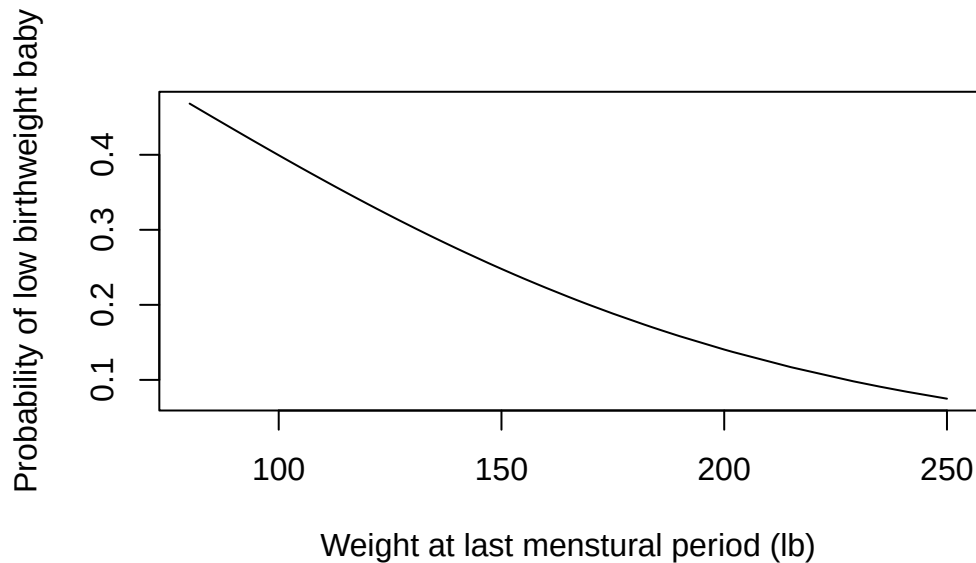
We might also display the fitted probabilities of a low birthweight baby by mother's weight in a graph. As usual in R there are lots of ways of doing this. Here, since there is a single covariate in the model, we could use:

```
plot(lbw$lwt, lwtMod$fitted.values,  
     xlab="Weight at last menstrual period (lb)",  
     ylab="Probability of low birthweight baby")
```



This plot could be improved - we may want to plot a smooth line through the fitted values rather than a scatter plot. One way we could do this is to create a data frame of the `lwt` covariate and fitted values, order it by `lwt`, and plot with `type="l"` argument (l for lines):

```
lwtModPlotData <- data.frame(lwt=lbw$lwt,
                             pihat=lwtMod$fitted.values)
#now order by lwt
lwtModPlotData <- lwtModPlotData[order(lwtModPlotData$lwt),]
plot(lwtModPlotData$lwt,
     lwtModPlotData$pihat,
     xlab="Weight at last menstrual period (lb)",
     ylab="Probability of low birthweight baby",
     type="l")
```



Question 3

We might prefer R to do the exponentiation process to convert to odds and odds ratios for us. Load (and install first if needed) the **broom** package. Examine the documentation for **tidy**. Then call **tidy** with your fitted logistic regression model, with suitable options to get odds and odds ratios and confidence intervals. Check your results by comparing with the ones you calculated by hand in the previous question.

SOLUTION

```
library(broom)
tidy(lwtMod, exponentiate=TRUE, conf.int=TRUE)
```

A tibble: 2 x 7

term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	2.71	0.785	1.27	0.205	0.617	13.6
2 lwt	0.986	0.00617	-2.28	0.0229	0.973	0.997

There are some slight differences in the CIs reported. This is because the **tidy** function calls the **confint** function, rather than **confint.default**. For a **glm** fit, **confint** calculates profile likelihood based confidence intervals, rather than the ± 1.96 SE ones (on the log odds scale).

Question 4

Using pen and paper and a calculator, calculate the probability that a women who weighs 120lbs (about 54.4kg) has a baby with birth weight less than 2500g.

SOLUTION

The estimated log odds of low birthweight for a mother who weighed 120 lbs is

$$(120 \times -0.014037) + 0.995763 = -0.6887$$

Hence, the probability is

$$\exp(-0.6887)/(1 + \exp(-0.6887)) = 0.334.$$

Question 5

Load (and install if necessary first) the `marginalEffects` package. Then use the `predictions` function with your earlier fitted model to check your answer to the previous question.

SOLUTION

The `predictions` function can create predicted values across a range of different values of the covariates in the model. Here we just want the predicted probability at a single value (120) of the sole covariate in the model. Therefore we can use:

```
library(marginalEffects)
predictions(lwtMod,
            newdata=datagrid(lwt=120))
```

```
lwt Estimate Pr(>|z|)      S 2.5 % 97.5 %
120      0.334   <0.001 15.5 0.268  0.408
```

Type: `invlink(link)`

This gives the same probability estimate as we obtained by hand. `prediction` here has used the fact that the fitted model is a logistic regression to know that it should calculate the estimate and 95% CI on the log odds scale, and then back-transform to the probability scale.

Question 6

Fit an appropriate model (using logistic or logit), and perform an appropriate test, to examine whether race has an association with the probability of having a low birthweight baby, conditional on (after adjusting for) mother's weight.

What do you conclude?

SOLUTION

To test this hypothesis, we need to compare the fit of our earlier model with one that adds race as an additional covariate. First we fit this model:

```
lwtRaceMod <- glm(low ~ lwt+race, data=lbw,  
                  family="binomial")  
summary(lwtRaceMod)
```

Call:

```
glm(formula = low ~ lwt + race, family = "binomial", data = lbw)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.802959	0.845059	0.950	0.3420
lwt	-0.015201	0.006438	-2.361	0.0182 *
raceBlack	1.080719	0.488013	2.215	0.0268 *
raceOther	0.481036	0.356646	1.349	0.1774

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 234.67 on 188 degrees of freedom
Residual deviance: 223.28 on 185 degrees of freedom
AIC: 231.28

Number of Fisher Scoring iterations: 4

Now we use the anova function:

```
anova(lwtMod, lwtRaceMod)
```

Analysis of Deviance Table

Model 1: low ~ lwt

Model 2: low ~ lwt + race

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	187	228.71			
2	185	223.28	2	5.4313	0.06616 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note that the function is called `anova`, since the models being passed to it are logistic regressions, not linear regressions, it is not ANOVA, but a likelihood-ratio (deviance) test that is being performed.

Comparing the fits of these models tests the null hypothesis that race has no independent association with the probability of having a low birthweight baby, after adjusting for mother's weight. The log-likelihood ratio statistic for this comparison is $-2 \times (-114.35403 - -111.63836) = 5.43$ which is compared to a chi-square distribution with two degrees of freedom, giving $p = 0.066$. There is some weak evidence that race is independently associated with low birthweight, conditional on mother's weight.

Part B: Investigation into effect of CS₂ at different doses on insect survival

The data are in `insecticide.RData`. There is one record per insect with `id` indexing the insect, `dosegroup` indicating which of eight dose groups the insect belonged to, `dose` giving the actual dose of insecticide received (mg/l) in the group and `y` indicating whether or not the insect died (0/1 equating to survived/died).

The aim of the analysis is to investigate the nature of the association between dose level and the proportion of insects killed.

Question 1

Load and explore the data. Are there any missing values? How many insects do you have data for?

SOLUTION

```
load("insecticide.RData")
summary(insecticide)
```

	id	dosegroup	dose	y
Min.	: 1	Min. :1.000	Min. :49.06	Min. :0.0000
1st Qu.:	121	1st Qu.:3.000	1st Qu.:56.91	1st Qu.:0.0000
Median	:241	Median :5.000	Median :64.76	Median :1.0000
Mean	:241	Mean :4.516	Mean :62.86	Mean :0.6008
3rd Qu.:	361	3rd Qu.:7.000	3rd Qu.:72.61	3rd Qu.:1.0000
Max.	:481	Max. :8.000	Max. :76.54	Max. :1.0000

```
nrow(insecticide)
```

```
[1] 481
```

```
table(insecticide$dose)
```

```

49.060001373291 52.9900016784668 56.9099998474121 60.8400001525879
              59              60              62              56
64.7600021362305 68.6900024414062 72.6100006103516 76.5400009155273
              63              59              62              60

```

There are no missing values. There are 481 insects, with approximately 60 insects in each of the dose groups.

Question 2

Fit a logistic regression model relating whether or not the insect died to **dose** treating this as a continuous variable. Is there evidence of an increasing risk of death with increasing dose?

SOLUTION

```

doseMod <- glm(y~dose, data=insecticide,
              family="binomial")
summary(doseMod)

```

Call:

```
glm(formula = y ~ dose, family = "binomial", data = insecticide)
```

Coefficients:

```
Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept) -14.0864      1.2284   -11.47   <2e-16 ***
dose         0.2366      0.0203    11.65   <2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 647.11  on 480  degrees of freedom
Residual deviance: 383.46  on 479  degrees of freedom
AIC: 387.46
```

Number of Fisher Scoring iterations: 5

The Wald test of the linear term has a z-score of 11.65, and $p < 0.001$. So we have strong evidence of increasing risk of death with increasing dose.

Question 3

Interpret the dose parameter in terms of an odds-ratio, and obtain its 95% CI.

SOLUTION

```
tidy(doseMod, exponentiate = TRUE,
     conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term          estimate std.error statistic  p.value    conf.low conf.high
<chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept) 0.000000763    1.23      -11.5 1.92e-30 0.0000000603 0.00000754
2 dose         1.27      0.0203     11.7 2.21e-31 1.22         1.32
```

There is strong evidence that the probability of death increases with increasing CS₂ dose ($p < 0.001$). Odds of death are estimated to increase by 27% (95% confidence interval 22% to 32%) per unit mg/l increase in dose.

Question 4

Use the fitted model to (manually) calculate the estimated probability of death at a dose of 55 mg/l CS₂.

SOLUTION

The estimated probability of death at a dose of 55 mg/l is given by the following.

$$\text{logit}(\hat{\pi}_i) = -14.0864 + (0.2366 \times 55) = -1.0734$$

and so

$$\hat{\pi}_i = e^{-1.0734} / (1 + e^{-1.0734}) = 0.254.$$

Question 5

Work out the dose that, on the basis of this fitted model, would lead to 50% of the insects dying (sometimes this is termed the LD50).

SOLUTION

The dose that, on the basis of this model, would lead to a 50% death rate (LD50) is the dose that is associated with $\pi = 0.5$ and hence $\text{logit}(\pi) = 0$. This can be found by solving the following equation.

$$0 = -14.084 + (0.2366 \times LD50)$$

Hence LD50 = 59.5 mg/l.

Question 6

Fit a logistic model that as well as `dose`, adds `I(dose^2)` as a covariate. Is there evidence that adding the quadratic effect of dose improves the fit? Finally, produce a plot of the fitted values from this new model, and also the fitted values from the earlier (linear model).

SOLUTION

```
doseQuadMod <- glm(y~dose+I(dose^2), data=insecticide,
                  family="binomial")
summary(doseQuadMod)
```

Call:

```
glm(formula = y ~ dose + I(dose^2), family = "binomial", data = insecticide)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.488178	9.867722	-0.252	0.801
dose	-0.150005	0.329179	-0.456	0.649

```
I(dose^2)      0.003187    0.002727    1.169    0.243
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 647.11  on 480  degrees of freedom
Residual deviance: 382.03  on 478  degrees of freedom
AIC: 388.03
```

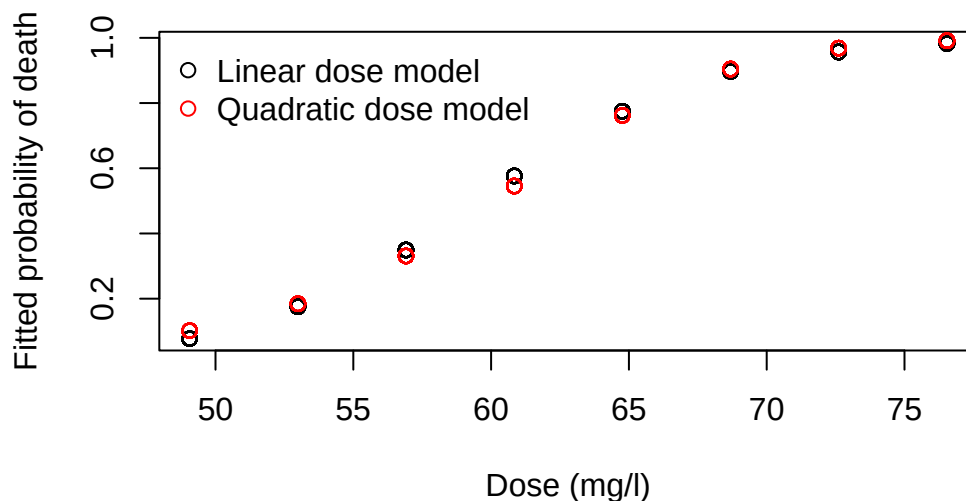
Number of Fisher Scoring iterations: 6

The p-value corresponding to the quadratic effect is $p = 0.243$ - there is no evidence of a quadratic effect of dose on the log odds of death.

Note that neither the linear or quadratic terms are statistically significant in the above model. This does not mean that we have no evidence of an association (indeed we have already seen that there is strong evidence of a linear association). One of these terms (but not both) can be removed from the model without making the fit worse. In the interests of simplicity we would usually drop the quadratic term.

We can plot the fitted probabilities from the two models using:

```
dosePlotData <- data.frame(dose=insecticide$dose,
                           pihat1=doseMod$fitted.values,
                           pihat2=doseQuadMod$fitted.values)
#now order by dose
dosePlotData <- dosePlotData[order(dosePlotData$dose),]
plot(dosePlotData$dose,
     dosePlotData$pihat1,
     xlab="Dose (mg/l)",
     ylab="Fitted probability of death")
points(dosePlotData$dose,
       dosePlotData$pihat2,
       col="red")
legend("topleft",
      legend = c("Linear dose model", "Quadratic dose model"),
      col     = c("black", "red"),
      pch     = c(1, 1),
      lty     = c(0, 0),
      bty     = "n")
```



We see the fitted values are very similar, which is consistent with the quadratic term not affecting the fit much. Note that for each model the predicted probabilities asymptotically approach 1 as the dose increases (and asymptotically approach 0 as the dose decreases).

Question 7

Now consider dose as a categorical/factor variable. Fit a further logistic model that treats dose as a factor, by using `factor(dose)` as a covariate.

Is the earlier model where we assumed a linear effect of dose (on the log odds of death) nested within the model you have just fitted?

SOLUTION

```
doseFactorMod <- glm(y~factor(dose), data=insecticide,
                     family="binomial")
summary(doseFactorMod)
```

Call:

```
glm(formula = y ~ factor(dose), family = "binomial", data = insecticide)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.1785	0.4307	-5.058	4.24e-07	***
factor(dose)52.9900016784668	0.8933	0.5327	1.677	0.0935	.
factor(dose)56.9099998474121	1.2847	0.5136	2.501	0.0124	*
factor(dose)60.8400001525879	2.1785	0.5069	4.298	1.73e-05	***

```

factor(dose)64.7600021362305    3.7319    0.5438    6.863 6.74e-12 ***
factor(dose)68.6900024414062    4.3571    0.6092    7.153 8.51e-13 ***
factor(dose)72.6100006103516    5.5797    0.8380    6.659 2.76e-11 ***
factor(dose)76.5400009155273    6.2561    1.0964    5.706 1.16e-08 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 647.11  on 480  degrees of freedom
Residual deviance: 378.84  on 473  degrees of freedom
AIC: 394.84

```

Number of Fisher Scoring iterations: 6

The constant is -2.179. This is the log odds of death in the baseline category (Group 1). Hence the probability is $e^{-2.179}/(1 + e^{-2.179}) = 0.10$.

The coefficient for the second dose category is 0.893. This is the log odds ratio comparing the second dose level with the first. Hence the log odds of death in the second level is

$$-2.179 + 0.893 = -1.285$$

It follows that the probability of death in the second dose group is $e^{-1.285}/(1 + e^{-1.285}) = 0.22$.

Interpretations for the other parameter estimates are analogous to this.

First note that in this factor or categorical dose model the predicted probabilities are the same as the empirically observed probabilities in each dose group. This is because dose only takes eight distinct values, so a model with eight parameters will give predicted probabilities that exactly match the observed group-specific probabilities. So, the model is effectively estimating the eight dose group probabilities of death with no constraints on these. The model with dose as continuous also gives just eight predictions (one for each dose) for the data at hand, but now the predicted probabilities are constrained such that they follow a linear (on the logit scale) relationship with dose. Hence this “continuous” dose model is nested in the “categorical” dose model.

Note that models where a predictor is treated as continuous are not nested in models where that predictor is categorised by grouping together similar (but not identical) levels of that predictor. For example, models where we i) consider age as continuous and ii) categorise age as “20-29”, “30-39” etc. are not nested.

Question 8

Carry out an appropriate likelihood ratio test to compare the fits of the models where dose is considered as continuous and categorical. What do you conclude?

SOLUTION

```
anova(doseMod,doseFactorMod)
```

Analysis of Deviance Table

Model 1: $y \sim \text{dose}$

Model 2: $y \sim \text{factor}(\text{dose})$

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	479	383.46			
2	473	378.84	6	4.6155	0.594

We have no evidence that the “categorical” dose model fits the data better than the “continuous” dose model. Again, if you produce a plot of the fitted probabilities from the two models in question, the differences in fitted values are fairly small (code and output not shown, but similar to that used earlier can be used again).